INCH-POUND
MIL-DTL-49285/31A
w/AMENDMENT 1
31 August 2018
SUPERSEDING
MIL-DTL-49285/31A
1 March 2010

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, ONE INDIVIDUALLY SHIELDED PAIR OF 22 (7 X 30) AWG

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

Engineering information: See table I.

Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.

TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	52 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	92 pF/ft
Continuous working voltage	300 V rms maximum between conductors
Shield dc resistance	14.0 ohms/1,000 feet maximum
Conductor dc resistance	18.5 ohms/1,000 feet maximum

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TABLE II. Description, physical.

Components	Construction details
Number of pairs	One individually shielded
Conductor type and wire size	Stranded tinned copper 22 AWG
Conductor stranding	7 X 30 AWG
Conductor insulation	Polyvinyl chloride
Conductor insulation thickness	0.016 inch nominal thickness
Drain wire type and size	Stranded tinned copper 22 AWG
Drain wire stranding	7 X 30 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.038 inch
Finished cable diameter	0.212 inch maximum outside diameter
Cable style (UL)	2464
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay length	9.6 twists per foot ± 10 percent

REQUIREMENTS:

Design and construction: (see table II):

Shield color code. The shield color shall be blue.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located between the shield and jacket, and is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 2556 flame test requirements.

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Shield integrity test.*

Frequency	Response
100 kHz	111 dB or more below reference
500 kHz	100 dB or more below reference
1 MHz	94 dB or more below reference
5 MHz	76 dB or more below reference
10 MHz	69 dB or more below reference
15 MHz	64 dB or more below reference
20 MHz	60 dB or more below reference
25 MHz	56 dB or more below reference
30 MHz	53 dB or more below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 37 dB below reference.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 97 $\pm$ 3.

Part or Identifying Number (PIN): The PIN shall be M49285/31.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-49285, this document references the following: UL 2556.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activity:
Army – MI
Air Force - 71
DLA – CQ

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